

CODE: 4. Healthcare Clinic Appointment Audit & No-Show Classifier

The image displays two screenshots of a Visual Studio Code editor window, showing the development of a Python script titled "4. Healthcare Clinic Appointment Audit & No-Show Classifier.py".

Top Screenshot: The script defines a dictionary of appointment data and uses pandas to create a DataFrame. It calculates the delay between the appointment and arrival times and classifies the status as "No Show Risk" or "On Time" based on the delay.

```
1 import pandas as pd
2 import numpy as np
3
4 data = {
5     "Patient": ["P1", "P2", "P3", "P4"],
6     "Appointment": [
7         "2025-06-01 09:00",
8         "2025-06-01 10:00",
9         "2025-06-01 11:00",
10        "2025-06-01 12:00"
11    ],
12     "Arrival": [
13         "2025-06-01 09:05",
14         "2025-06-01 10:30",
15         "2025-06-01 11:00",
16         "2025-06-01 12:45"
17    ]
18 }
19
20 df = pd.DataFrame(data)
21
22 df["Appointment"] = pd.to_datetime(df["Appointment"])
23 df["Arrival"] = pd.to_datetime(df["Arrival"])
24
25 df["Delay"] = (
26     df["Arrival"] - df["Appointment"]
27 ).dt.total_seconds() / 60
28
29 df["Status"] = np.where(
30     df["Delay"] > 20,
31     "No Show Risk",
32     "On Time"
33 )
34
```

Bottom Screenshot: The script is extended to print the DataFrame and the average delay.

```
35 print(df)
36
37 print("\nAverage Delay:")
38 print(df["Delay"].mean())

```

The screenshot shows a Jupyter Notebook with a Python script and its output. The script reads a CSV file and processes appointment data. The output is a table with the following data:

Patient	Appointment	Arrival	Delay	Status
0 P1	2025-06-01 09:00:00	2025-06-01 09:05:00	5.0	On Time
1 P2	2025-06-01 10:00:00	2025-06-01 10:30:00	30.0	No Show Risk
2 P3	2025-06-01 11:00:00	2025-06-01 11:00:00	0.0	On Time
3 P4	2025-06-01 12:00:00	2025-06-01 12:45:00	45.0	No Show Risk

Below the table, the script calculates the average delay:

```
Average Delay:
20.0
```

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